

Machine Health and Operational Efficiency Classification in 6G-Enabled Smart Manufacturing: A Logistic Regression Approach with Diagnostic Dashboard Deployment

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Abstract—Modern smart factories generate continuous streams of operational, sensor, production, quality, and network telemetry from interconnected, 6G-enabled industrial machines. While this Industrial Internet of Things (IIoT) data is collected at scale, manufacturing organizations frequently lack the analytical infrastructure to convert raw telemetry into actionable diagnostic intelligence regarding machine health, production efficiency, and network reliability. This paper presents an end-to-end diagnostic analytics framework applied to 100,000 operational records collected from 50 machines in a simulated 6G-enabled smart manufacturing environment. The pipeline integrates data validation, exploratory data analysis, correlation diagnostics, a manufacturing KPI engine, and a multinomial Logistic Regression classifier for predicting machine *Efficiency_Status* (High, Medium, Low) from eleven operational, sensor, quality, and network features. The trained model achieves an overall test accuracy of 91.71% on a stratified 80/20 split, with a weighted F1-score of 0.92, though performance is markedly stronger on the dominant Low-efficiency class ($F1 = 0.95$) than on the minority High-efficiency class ($F1 = 0.84$), reflecting the dataset’s natural class imbalance (77.8% Low, 19.2% Medium, 3.0% High). Correlation analysis reveals that the underlying sensor, production, quality, and network variables are almost entirely linearly independent of one another ($|r| < 0.02$ across all feature pairs), indicating that no single sensor channel dominates efficiency outcomes and that multivariate, model-based diagnostics are necessary rather than simple univariate thresholding. The complete analytical pipeline, trained model artifact, and a Streamlit-based interactive diagnostic dashboard supporting machine health monitoring, production diagnostics, quality analysis, 6G network performance tracking, cross-metric correlation exploration, and an executive KPI report are made available as a deployable analytics product. The study discusses the practical implications of class imbalance for predictive maintenance alerting, identifies the operational and network thresholds associated with reduced efficiency, and outlines a roadmap for extending the framework toward real-time, streaming, and explainable manufacturing intelligence systems.

Index Terms—Smart Manufacturing, Industrial Internet of Things, 6G Networks, Predictive Maintenance, Machine Health Monitoring, Operational Efficiency Classification, Logistic Regression, Manufacturing KPI Engineering, Class Imbalance, Diagnostic Dashboarding

I. INTRODUCTION

The convergence of Industry 4.0 automation, Industrial Internet of Things (IIoT) sensing, and emerging sixth-generation (6G) wireless communication is reshaping the operational fabric of modern manufacturing facilities. Where earlier generations of factory automation relied on isolated programmable logic controllers and batch-oriented reporting, contemporary smart factories continuously stream machine-level temperature, vibration, power, production throughput, predictive maintenance, quality, and network telemetry at sub-second resolution. 6G networks, with their projected sub-millisecond latency and near-zero packet loss targets, are positioned as the connective substrate that will enable real-time machine coordination, distributed control, and ultra-reliable low-latency industrial communication [1].

This proliferation of telemetry, however, creates a parallel challenge: the volume and dimensionality of operational data routinely exceed the analytical capacity of manual review or simple rule-based monitoring. A machine may simultaneously exhibit elevated vibration, marginally increased network latency, and a moderate predictive-maintenance score, none of which individually crosses an alarm threshold, yet the combination may be indicative of degraded operational efficiency. Converting this multivariate telemetry into an interpretable, actionable *efficiency classification* — and surfacing it through an operator-facing diagnostic interface — is the central problem this paper addresses.

This study analyzes 100,000 operational snapshots collected across 50 machines operating in three modes (Active, Idle, Maintenance) over a three-month observation window in a 6G-enabled smart manufacturing environment. Each record captures temperature, vibration, power consumption, network latency, packet loss, production speed, predictive maintenance score, error rate, and quality-control defect rate, alongside a derived three-class efficiency label. The specific contributions

of this work are as follows:

- A data validation and exploratory analysis pipeline that quantifies sensor range conformity, missing-value and duplicate-record integrity, and the (near-absent) linear correlation structure among manufacturing telemetry variables.
- A manufacturing KPI engine that computes a composite Machine Health Index, Defect Density Score, Error Frequency Index, and network-quality indicators directly from raw telemetry.
- A multinomial Logistic Regression classifier for *Efficiency_Status* prediction, trained on standardized operational, sensor, quality, and network features, evaluated with full per-class precision, recall, and F1 reporting to surface the practical impact of class imbalance.
- An interactive, multi-page Streamlit diagnostic dashboard — covering executive overview, machine health, production, quality, network, correlation analysis, and an executive report with live model inference — that operationalizes the analytical pipeline for plant-level decision support.
- A discussion of class-imbalance-aware deployment considerations and recommendations for extending the framework toward streaming, explainable, and predictive-maintenance-integrated manufacturing intelligence.

The remainder of this paper is organized as follows. Section II reviews related literature. Section III states the problem. Section IV defines the study objectives. Section V describes the dataset. Section VI details the methodology. Sections VII through XII present exploratory analysis, machine health, production, quality, network, and correlation diagnostics respectively. Section XIII presents the KPI engine. Section XIV describes the machine learning model and dashboard deployment. Section XV discusses results and limitations. Section XVI offers recommendations. Sections XVII and XVIII conclude the paper and outline future work.

II. LITERATURE REVIEW

A. Industrial IoT and Smart Manufacturing

The Industrial Internet of Things has been widely studied as the sensing backbone of Industry 4.0, enabling continuous machine-state monitoring and closed-loop process control [2]. Lee et al. introduced the Cyber-Physical Systems (CPS) framework for manufacturing, arguing that the integration of sensing, computation, and networking layers is a prerequisite for self-aware and self-predicting machine behavior [3]. The present study operationalizes this principle by treating sensor, network, and quality telemetry as a unified feature space for efficiency classification rather than analyzing each channel in isolation.

B. 6G and Ultra-Reliable Industrial Communication

Emerging 6G research emphasizes ultra-reliable low-latency communication (URLLC) as a foundational requirement for time-critical industrial automation, including closed-loop robotic control and synchronized multi-machine coordination

[1]. Network degradation — manifested as elevated latency or packet loss — has been identified as a contributing factor to coordination failures and reduced throughput in wirelessly networked manufacturing cells [4]. This motivates the explicit inclusion of network latency and packet loss as first-class diagnostic variables in the present framework, alongside conventional sensor and quality metrics.

C. Predictive Maintenance

Predictive maintenance research has established that combining vibration, temperature, and AI-derived health scores enables earlier and more accurate failure anticipation than fixed-interval maintenance schedules [5]. Susto et al. demonstrated that multi-sensor degradation modeling, rather than single-channel threshold alarms, substantially improves maintenance scheduling accuracy in semiconductor manufacturing [6]. The Machine Health Index constructed in this study follows this multivariate philosophy, combining normalized temperature, vibration, and predictive maintenance risk into a single composite indicator.

D. Manufacturing Efficiency and Quality Classification

Classification-based approaches to manufacturing efficiency and quality status have employed both linear and ensemble methods. Logistic Regression remains a widely used interpretable baseline in industrial classification tasks due to its coefficient-level interpretability and computational efficiency [7]. While ensemble and gradient-boosted methods frequently outperform linear baselines on structured tabular data [8], Logistic Regression continues to serve as both an interpretable production baseline and a useful diagnostic tool for understanding linear feature-outcome relationships before more complex models are deployed.

E. Class Imbalance in Industrial Classification

Operational efficiency, fault, and defect labels in real manufacturing settings are typically and structurally imbalanced, since machines spend the overwhelming majority of operating time in nominal or sub-optimal states rather than in clearly anomalous ones [9]. Studies on imbalanced industrial classification emphasize that overall accuracy can mask substantial degradation in minority-class recall, and recommend per-class precision/recall reporting alongside accuracy when evaluating fault- or efficiency-classification systems [10]. This concern is directly relevant to the present study, where the High-efficiency class constitutes only 3.0% of records.

F. Manufacturing Dashboards and Business Intelligence

Interactive dashboarding has been identified as a critical translation layer between analytical model output and operator decision-making in industrial settings [11]. Streamlit and similar rapid-development frameworks have gained adoption in applied data-science deployments for their ability to convert exploratory notebooks into production-facing interactive applications with minimal additional engineering overhead [12].

G. Research Gap

While substantial literature addresses IIoT sensing, predictive maintenance, and industrial classification individually, fewer studies present a single, reproducible pipeline that integrates sensor-, production-, quality-, and 6G-network-level diagnostics with a deployed, interactive decision-support dashboard grounded in an explicit class-imbalance-aware evaluation. The present study addresses this gap using a 100,000-record, 50-machine 6G-enabled manufacturing dataset.

III. PROBLEM STATEMENT

Smart manufacturing operators face a compounded analytical challenge: machine efficiency is jointly determined by sensor condition, production throughput, quality outcomes, and network reliability, yet each of these is typically monitored through separate, siloed dashboards or threshold alarms. This fragmentation prevents operators from identifying machines that are efficiency-degraded due to a *combination* of moderate factors rather than any single extreme reading, and it provides no systematic mechanism for prioritizing maintenance or operational intervention across a fleet of machines.

This study formalizes the problem along two axes:

Axis 1 — Machine Health and KPI Diagnostics: Given machine-level sensor, production, quality, and network telemetry, compute composite health and efficiency indicators (Machine Health Index, Defect Density Score, Error Frequency Index) and identify machines and operating conditions associated with degraded performance.

Axis 2 — Efficiency Classification: Given a feature vector $x = \{x_1, x_2, \dots, x_n\}$ comprising sensor, production, quality, maintenance, and network attributes, predict the categorical outcome $y \in \{\text{High, Medium, Low}\}$ representing the machine’s operational efficiency status, enabling proactive, fleet-wide operational monitoring.

IV. OBJECTIVES

The specific objectives of this study are:

- 1) To perform comprehensive data validation and exploratory data analysis of the 6G-enabled smart manufacturing dataset, including sensor range validation, missing-value and duplicate-record audits, and time-consistency checks.
- 2) To analyze machine-level sensor health across temperature, vibration, power consumption, and predictive maintenance score, and identify machines operating closest to operational thresholds.
- 3) To diagnose production performance, including top- and bottom-performing machines and production consistency across operation modes.
- 4) To analyze quality and error metrics, including defect rate and error rate distributions and their relationship to sensor and maintenance variables.
- 5) To evaluate 6G network performance through latency and packet-loss diagnostics across machines and operation modes.

- 6) To quantify the linear correlation structure among all numerical manufacturing variables and assess multi-collinearity risk for downstream modeling.
- 7) To engineer a manufacturing KPI engine producing a composite Machine Health Index and related diagnostic indicators.
- 8) To train and evaluate a multinomial Logistic Regression classifier for *Efficiency_Status* prediction, with explicit per-class performance reporting to surface class-imbalance effects.
- 9) To deploy the complete analytical pipeline as an interactive, multi-page Streamlit dashboard for plant-level decision support.

V. DATASET DESCRIPTION

A. Overview

The dataset comprises 100,000 operational records collected from 50 machines operating in a simulated 6G-enabled smart manufacturing environment over a three-month window (January–March), at minute-level granularity across hours 0–23. Each record represents a single machine-timestamp operational snapshot capturing sensor, production, quality, maintenance, and network state. The target variable *Efficiency_Status* classifies each record into High, Medium, or Low operational efficiency.

B. Feature Description

The dataset contains 19 columns spanning identification, sensor, production, quality, network, and temporal dimensions, summarized in Table I.

TABLE I
DATASET FEATURE DESCRIPTION

Feature	Type	Description
Machine_ID	Integer	Unique machine identifier (1–50)
Operation_Mode	Categorical	Active / Idle / Maintenance
Temperature_C	Float	Machine operating temperature (°C)
Vibration_Hz	Float	Machine vibration frequency (Hz)
Power_Consumption	Float	Electrical power consumed (kW)
Network_Latency	Float	6G network communication latency (ms)
Packet_Loss	Float	Network packet loss (%)
Production_Speed	Float	Production output (units/hr)
Maintenance_Score	Float	AI-generated predictive maintenance risk score (0–1)
Error_Rate	Float	Operational error rate (%)
Defect_Rate	Float	Quality-control defect rate (%)
Efficiency_Status	Categorical	Target: High / Medium / Low
Hour, Minute, Day, Month, Weekday	Temporal	Timestamp decomposition

C. Dataset Characteristics

Key descriptive statistics are summarized in Table II. The dataset exhibits zero missing values and zero duplicate records across all 100,000 entries, confirming high structural integrity suitable for direct modeling.

TABLE II
DATASET SUMMARY STATISTICS

Metric	Mean	Std. Dev.
Temperature (°C)	60.04	17.32
Vibration (Hz)	2.55	1.41
Power Consumption (kW)	5.75	2.45
Network Latency (ms)	25.56	14.12
Packet Loss (%)	2.49	1.44
Production Speed (units/hr)	275.92	130.10
Predictive Maintenance Score	0.499	0.289
Error Rate (%)	7.50	4.34
Defect Rate (%)	5.01	2.88

D. Class and Operation Mode Distribution

The *Efficiency_Status* target is structurally imbalanced: 77.8% of records are Low efficiency, 19.2% Medium, and 3.0% High. *Operation_Mode* is distributed as 70.1% Active, 20.1% Idle, and 9.9% Maintenance. This imbalance reflects realistic factory operation, in which the majority of operating time occurs under nominal-to-suboptimal conditions and sustained peak (High) efficiency is comparatively rare — a characteristic with direct consequences for classifier evaluation, discussed in Section XV.

VI. METHODOLOGY

A. Data Cleaning

An initial audit confirmed zero missing values and zero duplicate records across all 100,000 entries. Verbose source column names were standardized for downstream processing (e.g., `Production_Speed_units_per_hr` → `Production_Speed`, `Predictive_Maintenance_Score` → `Maintenance_Score`) without altering underlying values.

B. Outlier Detection

Outlier screening was performed using the Interquartile Range (IQR) method across all numerical sensor, production, quality, and network features, with boxplot visualization used to corroborate IQR-flagged observations. Given the bounded, sensor-realistic ranges of all variables (e.g., Temperature 30–90°C, Maintenance Score 0–1), flagged extreme values were retained as plausible operational states rather than removed, consistent with the principle that boundary-condition machine behavior is analytically informative rather than erroneous.

C. Correlation Analysis

A Pearson correlation analysis was conducted across all numerical features to assess multicollinearity risk prior to modeling. The resulting correlation matrix (Section XII) indicates that manufacturing sensor, production, quality, and network variables exhibit near-zero pairwise linear correlation ($|r| < 0.02$ in nearly all cases), with the only moderate relationship occurring between the temporal *Day* and *Month* fields ($r = -0.37$), an artifact of calendar structure rather than a manufacturing relationship.

D. KPI Engineering

A manufacturing KPI engine was constructed to derive composite operational indicators not directly observable from individual raw features, computed as:

Machine Health Index:

$$\text{MHI} = 100 \times \left(1 - \left(0.30 \hat{T} + 0.30 \hat{V} + 0.40 M_s \right) \right) \quad (1)$$

where $\hat{T}$ and $\hat{V}$ are min-max normalized Temperature and Vibration, and M_s is the Predictive Maintenance Score, treated as a 0–1 failure-risk indicator.

Defect Density Score: the mean Quality-Control Defect Rate (%) across the relevant machine or fleet scope.

Error Frequency Index: the mean operational Error Rate (%) across the relevant scope.

E. Feature Selection and Encoding

For the classification task, the feature matrix X was constructed from all operational, sensor, quality, maintenance, and network variables together with *Machine_ID*, *Operation_Mode*, and temporal fields (*Hour*, *Minute*, *Day*, *Month*), excluding the non-numeric *Date*, *Timestamp*, and *Weekday* columns. The categorical *Operation_Mode* feature and the *Efficiency_Status* target were label-encoded.

F. Train-Test Split and Feature Scaling

The dataset was partitioned using a stratified 80/20 train-test split (80,000 training, 20,000 test records), preserving the natural class proportions of *Efficiency_Status* in both partitions. All input features were standardized using a `StandardScaler` fitted exclusively on the training partition and subsequently applied to the held-out test partition, preventing data leakage.

G. Model

A multinomial Logistic Regression classifier was trained on the standardized feature set:

$$P(y = k \mid x) = \frac{e^{w_k^\top x + b_k}}{\sum_j e^{w_j^\top x + b_j}} \quad (2)$$

implemented via scikit-learn’s `LogisticRegression` with the LBFGS solver, a maximum of 1,000 iterations, and `random_state = 42` for reproducibility. The model, fitted scaler, and label encoders were serialized as a single artifact for downstream deployment in the diagnostic dashboard.

VII. EXPLORATORY DATA ANALYSIS

A. Sensor Distribution Patterns

Temperature, vibration, power consumption, network latency, and packet loss each exhibit approximately uniform distributions across their defined operational ranges (e.g., Temperature 30–90°C, Vibration 0.1–5.0 Hz), consistent with a simulated sensor-generation process spanning realistic full-range operating conditions rather than a process clustered tightly around a single nominal operating point.

B. Efficiency Status Distribution

The derived efficiency classification shows that the overwhelming majority of operational snapshots (77.8%) fall into the Low-efficiency category, with only 3.0% achieving High efficiency. This distribution implies that sustained high-efficiency operation is the exception rather than the norm across the observed fleet, underscoring the practical value of efficiency monitoring as a continuous diagnostic rather than a binary pass/fail check.

C. Operation Mode Distribution

Machines operate in Active mode for 70.1% of observed records, Idle for 20.1%, and Maintenance for 9.9%. Cross-tabulation of Operation Mode against Efficiency Status (Section XIV) provides the basis for understanding how operating context relates to predicted efficiency outcomes.

VIII. MACHINE-LEVEL SENSOR HEALTH

Machine-level aggregation across all 50 machines shows broadly consistent average sensor readings, with average Temperature, Vibration, Power Consumption, and Maintenance Score varying only modestly across machines, consistent with the near-uniform, independent feature-generation process identified in the correlation analysis. Machine-level Health Index scores, computed via Equation (1), were used to rank machines and surface those operating closest to combined sensor and maintenance-risk thresholds for prioritized inspection, as implemented in the Machine Health page of the deployed dashboard.

IX. PRODUCTION PERFORMANCE DIAGNOSTICS

Average production speed across the fleet is 275.92 units/hr (SD = 130.10), reflecting substantial throughput variability across machines and operating conditions. Top- and bottom-decile machine rankings by average production speed were computed to identify both best-practice operating profiles and machines warranting throughput investigation. Production speed consistency, measured via within-machine and within-operation-mode standard deviation, is highest for machines in Active mode and substantially reduced for machines in Idle and Maintenance modes, as expected given the operational semantics of these states.

X. QUALITY AND ERROR ANALYSIS

Mean operational Error Rate (7.50%) and mean Quality-Control Defect Rate (5.01%) were analyzed jointly to identify quality bottlenecks. Scatter and distributional analysis (implemented in the Quality dashboard page) examined the relationship between Temperature and Defect Rate, and between Vibration and Error Rate; consistent with the global correlation findings (Section XII), no strong linear relationship was observed between these sensor and quality variables in aggregate, indicating that quality degradation in this environment is not attributable to any single dominant sensor channel but instead reflects the joint, largely independent combination of operational factors captured by the multinomial classification model.

XI. 6G NETWORK PERFORMANCE DIAGNOSTICS

Average network latency across the fleet is 25.56 ms (SD = 14.12) and average packet loss is 2.49% (SD = 1.44), both well within typical Industrial IoT operational tolerances under non-critical 6G slicing, though individual machine-hours exceeding these averages were flagged for SLA-risk monitoring in the dashboard's Network page. Hourly aggregation of latency and packet loss was used to identify whether time-of-day network congestion patterns exist; consistent with the broader correlation findings, no pronounced systematic hourly degradation pattern was observed in this dataset.

XII. CROSS-METRIC CORRELATION DIAGNOSTICS

A. Correlation Matrix

A full Pearson correlation matrix was computed across all nine continuous operational variables plus temporal fields. With the sole exception of the expected Day-Month relationship ($r = -0.37$), all pairwise correlations among Temperature, Vibration, Power Consumption, Network Latency, Packet Loss, Production Speed, Maintenance Score, Error Rate, and Defect Rate fall below $|r| = 0.02$.

B. Interpretation

This near-total linear independence among manufacturing telemetry variables carries two practical implications. First, multicollinearity is not a concern for the Logistic Regression classifier, and no feature elimination was required prior to modeling. Second — and more substantively — it implies that machine efficiency in this environment cannot be reliably inferred from any single sensor channel or simple bivariate threshold rule; efficiency status is a genuinely multivariate function of the joint configuration of sensor, production, quality, maintenance, and network state, which directly motivates the use of a multivariate classification model rather than univariate alerting rules.

XIII. MANUFACTURING KPI ENGINE

The KPI engine (Section VI-D) computes, for any filtered scope of the fleet (factory-wide, per-machine, or per-operation-mode):

- Machine Health Index (Equation (1))
- Average Production Speed
- Defect Density Score
- Error Frequency Index
- Average Network Latency and Packet Loss
- Average Maintenance Score, Temperature, Vibration, and Power Consumption

These indicators are surfaced through KPI cards, gauge charts, and an executive KPI table in the deployed dashboard, providing a continuously updated, filter-aware operational summary for plant managers.

XIV. MACHINE LEARNING MODEL AND DASHBOARD DEPLOYMENT

A. Model Performance

The Logistic Regression classifier achieved an overall test accuracy of **91.71%** on the stratified 20,000-record held-out test set. Table III reports full per-class precision, recall, and F1-score.

TABLE III
LOGISTIC REGRESSION CLASSIFICATION REPORT (TEST SET, N = 20,000)

Class	Precision	Recall	F1	Support
High	0.86	0.83	0.84	597
Medium	0.80	0.76	0.78	3,838
Low	0.95	0.96	0.95	15,565
Accuracy		0.9171		20,000
Macro Avg	0.87	0.85	0.86	20,000
Weighted Avg	0.92	0.92	0.92	20,000

The model performs strongly on the majority Low-efficiency class (F1 = 0.95) and reasonably on the High-efficiency class (F1 = 0.84) despite its small support (597 of 20,000 test records, 3.0%), while the Medium class — which sits at the decision boundary between Low and High behavior — shows the lowest precision and recall (0.80 / 0.76), consistent with the expected difficulty of separating an intermediate class in an ordinal three-class problem using a linear decision boundary.

B. Dashboard Deployment

The complete analytical pipeline is deployed as a multi-page Streamlit application comprising:

- **Dashboard** — executive KPI cards, factory health gauges, efficiency-status trend and mix visualizations, and quick insights.
- **Machine Health** — data validation summary, per-machine sensor averages, radar chart, machine×hour heatmap, and health scorecards highlighting machines closest to operational thresholds.
- **Production** — top/bottom-10 machine rankings, production trend, consistency by operation mode, and distribution analysis.
- **Quality** — error and defect scatter/box plots, histograms, machine quality rankings, and correlation diagnostics.
- **Network** — latency vs. packet-loss diagnostics, hourly network-quality trend, and SLA-risk tracking.
- **Correlation Analysis** — cross-metric scatter plots, full correlation heatmap, interactive pair plot, and an operation-mode→efficiency sunburst.
- **Executive Report** — the KPI engine table together with a live inference panel that loads the serialized Logistic Regression artifact (model, scaler, and label encoders) to classify user-specified or filtered-fleet operating conditions in real time, including class-probability visualization and a confusion matrix evaluated on the currently filtered data sample.

Sidebar filters (Machine ID, Operation Mode, Month, Day, Hour range, primary metric, and multi-feature selection) apply consistently across all pages, enabling drill-down analysis from factory-wide KPIs to individual-machine diagnostics within a single interactive session.

XV. RESULTS AND DISCUSSION

A. Class Imbalance and Evaluation Validity

The headline 91.71% accuracy figure, while respectable for a linear baseline, must be interpreted in light of the dataset’s 77.8%/19.2%/3.0% class distribution: a naive classifier that always predicts *Low* would achieve approximately 77.8% accuracy without learning any structure. The Logistic Regression model’s weighted F1-score of 0.92 and macro-averaged F1-score of 0.86 indicate genuine discriminative capability beyond the majority-class baseline, particularly given that the minority High-efficiency class (3.0% of test data) is recovered with 0.83 recall — meaning the model correctly identifies the large majority of genuinely high-efficiency operating states despite their rarity. Nonetheless, the gap between the Low-class F1 (0.95) and the Medium- and High-class F1 scores (0.78 and 0.84) indicates that production deployment of this model for automated alerting should weight per-class performance, not aggregate accuracy, when setting confidence thresholds for operator notification.

B. Implications of Near-Zero Feature Correlation

The near-total absence of linear correlation among sensor, production, quality, and network variables (Section XII) is a notable structural property of this environment. Practically, it means that no individual telemetry channel can serve as a reliable proxy for overall machine efficiency, reinforcing the case for multivariate, model-based classification over single-metric threshold alarms as the basis of the diagnostic dashboard’s efficiency monitoring. It also suggests that linear models such as Logistic Regression are reasonably well-matched to this feature space, since the absence of strong nonlinear interaction signals evidenced by simple correlation screening means that the marginal benefit of substantially more complex ensemble models may be limited — a hypothesis that should be empirically tested in future work (Section XVIII).

C. Operational Interpretation

The combination of a 91.71%-accurate efficiency classifier and a real-time KPI engine provides plant operators with two complementary diagnostic layers: (1) a continuously updated, interpretable summary of factory-wide and machine-level health and throughput indicators, and (2) a probabilistic efficiency classification that can flag machines transitioning toward Medium or Low efficiency states before they manifest as overt production or quality failures, supporting the predictive-maintenance and operational-intervention use cases outlined in Section I.

D. Limitations

This study has several limitations. First, the three-month observation window limits the ability to assess seasonal or longer-term degradation trends. Second, the near-zero inter-feature correlation, while analytically convenient, may reflect characteristics of the data-generation process and should be validated against real plant-floor telemetry before the framework is deployed in a live industrial setting. Third, only a single linear model (Logistic Regression) was evaluated in this iteration of the study; a comparative evaluation against ensemble methods (Random Forest, XGBoost, LightGBM) — as has proven valuable in adjacent profitability- and fault-classification domains — was intentionally deferred to future work to first establish an interpretable, deployable baseline.

XVI. RECOMMENDATIONS

A. Class-Imbalance-Aware Alerting

Given the Medium-class F1-score of 0.78, automated dashboard alerts should incorporate prediction-probability thresholds rather than relying solely on the predicted class label, and should be validated against domain-expert review before triggering maintenance dispatch.

B. Machine Health Index as a Leading Indicator

The composite Machine Health Index (Equation (1)) should be tracked as a continuous leading indicator alongside the discrete *Efficiency_Status* classification, since composite scores expose gradual degradation trends that a three-class label may not surface until a class boundary is crossed.

C. Comparative Model Evaluation

Future iterations of the production model should benchmark Logistic Regression against Random Forest, XGBoost, and LightGBM classifiers using the same standardized feature set and stratified evaluation protocol, reporting per-class metrics consistently with Table III to determine whether nonlinear ensemble methods materially improve Medium- and High-class recall.

D. Network-Aware Maintenance Scheduling

Given the explicit inclusion of 6G network latency and packet loss as model features, maintenance and production scheduling should account for network-quality state alongside conventional sensor and quality metrics, particularly for machines operating in coordinated, network-dependent automation cells.

E. Dashboard-Driven Operational Reviews

The deployed Streamlit dashboard's Executive Report page should be incorporated into routine plant management review cycles, using the live confusion matrix and KPI table evaluated on the current operational filter to track model performance drift and operational trends over time.

XVII. CONCLUSION

This paper has presented an end-to-end diagnostic analytics framework for machine health and operational efficiency monitoring in a 6G-enabled smart manufacturing environment. Applied to 100,000 operational records from 50 machines, the framework integrates data validation, exploratory analysis, correlation diagnostics, a manufacturing KPI engine, and a multinomial Logistic Regression classifier achieving 91.71% test accuracy and a weighted F1-score of 0.92 for three-class efficiency prediction, deployed through an interactive, multi-page Streamlit dashboard.

The central analytical findings are twofold. First, manufacturing sensor, production, quality, and network variables in this environment exhibit near-total linear independence, indicating that machine efficiency cannot be reliably inferred from any single telemetry channel and must instead be assessed through multivariate, model-based diagnostics. Second, the natural 77.8%/19.2%/3.0% class imbalance of *Efficiency_Status* means that aggregate accuracy alone understates the practical evaluation requirements of a deployed efficiency classifier; per-class precision and recall — particularly for the rare but operationally important High-efficiency class — must guide both model selection and alerting-threshold design.

The deployed dashboard operationalizes these findings into a decision-support tool spanning executive KPIs, machine health, production, quality, network, and correlation diagnostics, with live model inference embedded directly into the executive reporting workflow, providing a practical foundation for predictive-maintenance-integrated, 6G-aware smart manufacturing operations.

XVIII. FUTURE SCOPE

Comparative Ensemble Modeling: Future work should benchmark Random Forest, XGBoost, and LightGBM classifiers against the Logistic Regression baseline established here, with explicit attention to Medium- and High-class recall improvement.

Class Imbalance Mitigation: Techniques such as class-weighted loss functions, SMOTE-based oversampling, or threshold-moving calibration should be evaluated to improve minority-class (High-efficiency) performance without degrading majority-class precision.

Temporal and Streaming Modeling: Extending the three-month cross-sectional dataset to a longitudinal or streaming setting would enable trend-based degradation forecasting (e.g., LSTM or temporal convolutional models) rather than point-in-time classification, more directly supporting predictive maintenance use cases.

Explainability Integration: Incorporating SHAP or LIME-based explanation into the live dashboard inference panel would allow operators to understand which specific features drove a given efficiency prediction for an individual machine, improving trust and actionability.

Real Plant-Floor Validation: The near-zero inter-feature correlation observed in this dataset should be validated against

real industrial telemetry, since genuine plant-floor data may exhibit stronger sensor-to-sensor and sensor-to-quality coupling that would materially affect both model architecture choice and KPI engineering.

Anomaly Detection Layer: Complementing the supervised efficiency classifier with an unsupervised anomaly-detection layer (e.g., Isolation Forest or autoencoder reconstruction error) would enable detection of genuinely novel failure modes not represented in the historical *Efficiency_Status* label distribution.

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